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Manas Sashank Juvvi

Profile

Robotics and Control Engineer with an MSc in Systems and Control from TU Delft and over 1.5 years of full-time research experience. Experienced in control algorithm design, computer vision, perception, motion planning and system modelling for mobile robots, UAVs and robotic manipulators. Hands-on experience developing and testing robotic systems from simulation to real-world deployment, with proficiency in C++, Python, MATLAB, ROS 2, Gazebo, NVIDIA Isaac Sim and Isaac Lab. Strong background in integrating control software with the mechanical and electronic components of robotic systems.

Education

2024 – 2026 *MSc Systems and Control* — *Delft University of Technology*

Thesis (Nov 2025 – Aug 2026): *A Statistical Framework for Counterexample Driven Refinement of Reinforcement Learning Controllers*, supervised by Dr. Manuel Mazo Espinosa.

- Developed a *train-certify-refine* framework for improving *reinforcement learning controllers* for mobile robots under task and *safety constraints*, combining controller training, refining and *statistical certification* from finite rollout data. Built and validated in *NVIDIA Isaac Lab*.
- Achieved *44% higher mean task robustness* than the leading baseline (TGPO*) with *4× lower compute time*.

Coursework: Model Predictive Control, Modeling of Dynamical Systems, Optimization, Machine Learning for Systems and Control, Autonomous Flight of Micro Air Vehicles, Control in Human-Robot Interaction

Jun 2025 *ETH Zürich Robotics Summer School* — *ETH Zürich, Geneva*

Selected for a competitive robotics summer school, where I worked as part of a team to develop and deploy the full autonomy stack of a rough-terrain mobile robot, integrating *perception, localisation, mapping, planning and navigation* using a *Docker-based development and deployment workflow*.

2019 – 2023 *BTech Mechatronics*, minor in Robotics and Automation — *Manipal Institute of Technology*

CGPA 8.48/10 ▪ *Coursework*: Robot Dynamics and Control, Linear Control Theory, Mobile Robots and Path Planning, Machine Vision and Image Processing, Artificial Intelligence

Experience

Jan 2023 – Jul 2024 *Research Associate*, Robert Bosch Centre for Cyber-Physical Systems — *Bangalore, India*

Formal Control and Autonomous Systems (FOCAS) Lab, PI Dr. Pushpak Jagtap

- Designed and built a *research-grade outdoor mobile robot* from concept to field-tested prototype, covering *electronics, embedded software, mechanical assembly and autonomous navigation*.
- Developed and maintained *C++ and Python robotics software* using *ROS 2* across three platforms (TurtleBot3 fleet, F1TENTH race car, outdoor rover).
- Carried out two research projects on *safe control and planning* from problem formulation through algorithm development, simulations and hardware deployment, resulting in first-author papers at *ICRA 2024* and *IROS 2025*.
- Mentored 4 junior students on ROS, reinforcement learning and formal control methods.

Selected Projects

Jan 2023 – Jun 2024 *Research-Grade Outdoor Mobile Robot* | *OAK-D, Jetson Xavier NX, C++, Python, ROS 2, Arduino*

- Developed and deployed the *perception pipeline* using an *OAK-D depth camera* for terrain classification and object-triggered behaviour, running onboard a *Jetson Xavier NX*.
- Developed *low-level control algorithms* using *prescribed-performance control* for accurate trajectory tracking, implemented as *embedded control software* on the robot, and integrated *ROS packages for autonomous navigation and exploration*.

Jun 2023 – Jun 2024 — *F1TENTH Autonomous Race Car* | *C++, Python, ROS 2, PyTorch, MATLAB, Gazebo*

- Developed a *physics-informed, data-driven Gaussian-process model* for improved vehicle-state prediction accuracy at high speeds.
- Developed a *dynamic-obstacle prediction* module that evaluates collision risk across the planner's state space, enabling more aggressive trajectory generation in low-risk regions.

Sep 2023 – Mar 2024 — *Motion Primitive Reinforcement Learning* | *Python, PyTorch, ROS 2*

- Combined *reinforcement learning and sampling-based motion planning* to adapt robot behaviour to new task specifications *without retraining*.
- Developed a *Gaussian-process reachability model* linking planning and control. Validated the approach on *TurtleBot3 and Unitree Go1* in simulation and on hardware.

Apr 2023 – Sep 2023 — *Safe Multi-Robot Exploration* | *C++, Python, ROS 2, multi-agent systems*

- Developed a multi-robot exploration stack with provable collision avoidance, using symbolic control for each robot's local controller and control barrier functions to compose them safely.
- Designed and deployed a *frontier target selection algorithm* that ranks exploration targets using safety margins around unmapped obstacles on a *TurtleBot3 fleet*.

Sep 2021 – Sep 2022 — *e-Yantra Robotics Competition: Strawberry Stacker* ↗ | *Python, OpenCV, ROS, PX4* video ↗

- Built a *vision-guided pick-and-place* system for multiple quadcopters with *ArUco marker detection* to locate boxes and *colour/contour-based detection* to locate trucks for loading.
- Fed detected positions into the *control and navigation pipeline* (ROS, PX4) to guide the drones to pick and place targets. Team reached the *Grand Finals (top 8 of 250+ teams)*.

Publications

IROS 2025 — *Signal Temporal Logic Compliant Co-design of Planning and Control* ↗
video ↗ — **M. S. Juvvi**, T. D. Kurne, V. J, S. Kolathaya, P. Jagtap
IEEE/RSJ International Conference on Intelligent Robots and Systems, Hangzhou, China

ICRA 2024 — *Safe Multi-Robot Exploration using Symbolic Control* ↗
video ↗ — **M. S. Juvvi**, D. S. Sundarsingh, R. Das, P. Jagtap
IEEE International Conference on Robotics and Automation, Yokohama, Japan

Skills

Languages — C++ (2+ years), Python (3+ years), MATLAB & Simulink, CUDA, Embedded C, Bash

Tools & Frameworks — ROS 2, ROS, PyTorch, Docker, Linux, Gazebo, NVIDIA Isaac Sim & Isaac Lab, Git, CMake, OpenCV, TensorFlow, PX4, ABB RobotStudio

Domain Expertise — Software architecture & system design, perception & sensor processing, machine learning model integration, autonomous navigation, control system design, motion & trajectory planning, model predictive control, dynamic modelling & simulation, system identification, state estimation, sensor fusion, safety-critical control, reinforcement learning, SLAM

Hardware Platforms — *Embedded*: Jetson Xavier NX, Raspberry Pi, ESP32 ▪ *Mobile & aerial*: TurtleBot 2/3/4, F1TENTH, Unitree Go1, Crazyflie 2.1 ▪ *Manipulators*: ABB IRB 2600, Franka Emika Panda, UR5e ▪ *Sensors*: OAK-D depth camera, IMU, LIDAR

Extracurricular

Jun 2021 – Oct 2022 — *Manipal Institute of Technology* — *Manipal, India*
Vice Chairperson, IEEE Robotics and Automation Society Student Chapter

- Grew the chapter to 50+ active members and organised an industrial robotics lecture series.
- Compiled educational resources covering programming, robot kinematics, ROS, and machine learning for new club members.

Jul 2021 – Sep 2022 — *Manipal Institute of Technology* — *Manipal, India*
General Secretary, IE Mechatronics Students' Chapter

- Founded the Robotics subsystem and mentored new members in programming, linear algebra, robot dynamics and ROS.
- Organised "Robotron," a hands-on workshop on the UR5e and TurtleBot 2 platforms.